Network Security Overview

- Confidentiality, integrity, availability
- Threat model: Dolev-Yao attacker
- Agenda for today

TCP Three-Way Handshake

- SYN →
- ← SYN-ACK
- ACK →
- Connection established

Key Takeaways

- Always validate input
- Use TLS 1.3
- Rotate keys regularly

Comparison

Option A

Fast

Reliable

Expensive

Option B

Cheap

Simple

Slow